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PROJECT PROPOSAL ON

BloodLink

SUBMITTED BY

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SUBMITTED TO

Department of Computer Engineering

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Project Title

BloodLink

Aim of the project

To build a platform that connect blood donor and blood receiver.

Abstract

The proposal highlights the efficient and user-friendly nature of our app, simplifying the easy communication between blood receiver and blood donor. By leveraging advanced technology and a network we provide users with a hassle-free solution to receive a required blood. Provide users with a hassle-free solution to obtain their required medications. The proposal emphasizes the benefits of our system, including convenience, easy access and maintaining privacy of the user's data.

INTRODUCTION

This idea focuses on allowing users to upload their basic information to create an account from which a user can requests blood from locally available blood donors of same blood group. This system will locate the donors using google location or the user can manually search blood donors of a certain location. In order to donate the blood by a user the user need to create a profile which can be created as soon as user sign up or can create later. Users profile contains user's name, address, phone number, email, blood group, date of birth and last time of blood donation. As soon as a requester request a blood the system automatically sends the notification to all the donors who are available nearby of the receiver. The donor has features to accept or reject the request requested by the receiver. As soon as the donor accepts the receiver's request the receiver can message donor and see the contact details in their profiles. Without the donor access others can't see any personal details such as phone number and email of the donor.

OBJECTIVES

Every project must have some objectives that lead to the development of the project. Our project also has some objectives that must be fulfilled by the project.

- To provide convenience platform where blood donor and blood receiver can easily communicate.

EXISTING SYSTEM

In the existing system receiver needs to visit the Nepal Red Cross Society to receive the blood however, it is difficult to find the required blood on time and it is costly sometimes. Also required blood might not be available sometimes. Moreover, it is hard to find same blood group donors.

FEASIBILITY STUDY

A feasibility study is an analysis that takes all of a project's relevant factors into account including economics, technical and scheduling considerations to ascertain the likelihood of completing the project successfully.

TECHNICAL FEASIBILITY

- Infrastructure: Assess the availability and adequacy of the required technical infrastructure, including servers, databases, and internet connectivity, to support the app and handle user interactions and user requests.
- App Development: Determine the technical expertise and resources needed for app development, ensuring compatibility across various devices and operating systems.
- Security: Address the technical measures required to ensure the privacy and security of user personal data.

Economic Feasibility:

- Cost savings by eliminating the need for physical infrastructure, transportation, and manpower for traditional blood donation events.
- Efficient resource utilization by connecting recipients directly with nearby donors, reducing wastage and ensuring optimal distribution.
- Time and effort reduction by automating the matching process between donors and recipients based on blood type and location.
- Increased availability of blood by expanding the donor pool and improving access for those in need.
- Positive social impact by enhancing the well-being of individuals and communities, indirectly contributing to reduced healthcare costs and increased productivity.

Operational Feasibility:

- User-friendly interface for easy adoption and navigation.
- Scalability to handle increasing user base and demand.
- Integration with location services for efficient matching based on proximity.
- Robust database management system for secure storage and retrieval of user information.
- Privacy and security measures to protect sensitive data and ensure compliance with regulations.

Legal and Regulatory Feasibility:

- Data Protection: Compliance with Nepal's Data Protection Act and other relevant regulations to ensure the privacy and security of user data, including obtaining appropriate consent for data collection and defining data retention and disposal policies.

- Health and Medical Regulations: Adherence to Nepal's laws and regulations related to blood donation, handling of personal health information, and any specific requirements set forth by relevant authorities.
- Jurisdictional Compliance: Ensuring compliance with all applicable laws, regulations, and guidelines specific to Nepal, including those related to data protection, health regulations, electronic communications, and any other relevant legal requirements established by the government or regulatory bodies in Nepal.

ADVANTAGES

- Enhanced access to blood by connecting recipients with nearby donors, reducing response time during emergencies.
- Efficient resource utilization by optimizing the distribution of blood, reducing wastage, and improving availability for those in need.
- Time and effort savings through automated matching of donors and recipients, eliminating manual searching and contacting.
- Quick notification and response system, enabling real-time communication and prompt donor engagement.
- Potential to save lives by facilitating timely access to blood and creating a network of willing donors, with positive social impact and cost reduction as additional benefits.

FUTURE ENHANCEMENT

- Open our own blood bank.
- Donor health monitoring system.

CONCLUSION

In summary, our BloodLink project is focused on efficiently connecting blood donors with recipients. The system facilitates the process through user accounts, local donor searches, and blood requests, ensuring improved access to life-saving blood transfusions. Donor profiles prioritize privacy and security. Future enhancements include establishing our own blood bank and implementing a donor health monitoring system to enhance effectiveness and sustainability. Our goal is to have a positive impact on the lives of those in need by maintaining control over the blood supply chain and prioritizing donor health.

Approved By

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